

1 Solution — GIAM 7.1

Problem 8

1.1 Problem

How many subsets does a finite set have?

1.2 Setup

Let S be a finite set with $|S| = n$ elements, $n \geq 0$. Denote by $\mathcal{P}(S)$ the *power set* of S — the set of all subsets of S :

$$\mathcal{P}(S) = \{A : A \subseteq S\}.$$

Both the empty set $\emptyset$ and the set S itself count as subsets. We want $|\mathcal{P}(S)|$.

1.3 Claim

$$\boxed{|\mathcal{P}(S)| = 2^n}.$$

We give three proofs — by the multiplication principle, by summing over subset sizes, and by induction — each illuminating a different perspective.

1.4 Proof 1 — Element-by-Element Decisions

Specifying a subset $A \subseteq S$ amounts to making, *independently for each element of S* , a binary decision: “is this element in A , or not?” With n elements and 2 choices each, the multiplication principle gives $2 \cdot 2 \cdots 2 = 2^n$ subsets in total.

Formally, write $S = \{s_1, s_2, \dots, s_n\}$ and define the *characteristic-function* map

$$\chi: \mathcal{P}(S) \longrightarrow \{0,1\}^n, \quad \chi(A) = (b_1, b_2, \dots, b_n), \quad b_i = \begin{cases} 1 & \text{if } s_i \in A, \\ 0 & \text{otherwise.} \end{cases}$$

This χ is a bijection — its inverse reads a binary n -tuple as the indicator of which elements to include. Hence

$$|\mathcal{P}(S)| = |\{0,1\}^n| = 2^n. \quad \blacksquare$$

1.5 Proof 2 — Summing Over Subset Sizes

Partition $\mathcal{P}(S)$ by subset size:

$$\mathcal{P}(S) = \bigsqcup_{k=0}^n \mathcal{P}_k(S), \quad \mathcal{P}_k(S) = \{A \subseteq S : |A| = k\}.$$

The standard count gives $|\mathcal{P}_k(S)| = \binom{n}{k}$. Summing and applying the *binomial theorem* with $x = y = 1$,

$$|\mathcal{P}(S)| = \sum_{k=0}^n \binom{n}{k} = \sum_{k=0}^n \binom{n}{k} 1^k 1^{n-k} = (1+1)^n = 2^n. \quad \blacksquare$$

1.6 Proof 3 — Induction on n

Base case ($n = 0$). The only set of size 0 is the empty set $\emptyset$, whose only subset is itself: $|\mathcal{P}(\emptyset)| = 1 = 2^0$. ✓

Inductive step. Suppose $|\mathcal{P}(T)| = 2^n$ for every set T with $|T| = n$. Let S have $|S| = n + 1$, fix any element $a \in S$, and let $T = S \setminus \{a\}$, so $|T| = n$.

Every subset of S either contains a or does not, partitioning $\mathcal{P}(S)$ into two disjoint pieces:

- **Subsets *not* containing a .** These are exactly the subsets of T , so there are $|\mathcal{P}(T)| = 2^n$ of them by the inductive hypothesis.
- **Subsets *containing* a .** Each such subset has the form $A \cup \{a\}$ for a unique $A \subseteq T$, and the assignment $A \mapsto A \cup \{a\}$ is a bijection between $\mathcal{P}(T)$ and the family of subsets of S containing a . Again 2^n .

Adding,

$$|\mathcal{P}(S)| = 2^n + 2^n = 2 \cdot 2^n = 2^{n+1}.$$

By induction, $|\mathcal{P}(S)| = 2^{|S|}$ for every finite set S . ■

1.7 Sanity Check on Small Cases

n	S	$\mathcal{P}(S)$	2^n
0	$\emptyset$	$\{\emptyset\}$	1
1	$\{a\}$	$\{\emptyset, \{a\}\}$	2
2	$\{a, b\}$	$\{\emptyset, \{a\}, \{b\}, \{a, b\}\}$	4
3	$\{a, b, c\}$	$\{\emptyset, \{a\}, \{b\}, \{c\}, \{a, b\}, \{a, c\}, \{b, c\}, \{a, b, c\}\}$	8

Table 1.1

All entries agree with 2^n .

1.8 Connection — The Cryptographer’s Alphabet

Taking $S = \Sigma = \{\mathbf{a}, \mathbf{b}, \mathbf{c}, \mathbf{d}, \mathbf{e}, \mathbf{f}\}$ (the alphabet from Problems 5–7), Proof 1 gives $|\mathcal{P}(\Sigma)| = 2^6 = 64$ subsets. Proof 2 recovers the same total by summing the row of Pascal’s triangle for $n = 6$:

k	0	1	2	3	4	5	6	Total
$\binom{6}{k}$	1	6	15	20	15	6	1	64

Table 1.2

The $\binom{6}{4} = 15$ entry is precisely the count of 4-element subsets — i.e., the 15 increasing-order words found in Problem 6. The symmetric row reflects the identity $\binom{6}{k} = \binom{6}{6-k}$ from Problem 4.

1.9 Remark — Why “Power Set”

The traditional name *power set* is no accident: the cardinality $|\mathcal{P}(S)| = 2^{|S|}$ is literally a *power of two*. The notation 2^S is sometimes used in place of $\mathcal{P}(S)$ for the same reason — both as a mnemonic for the cardinality and because of the bijection χ of Proof 1 between $\mathcal{P}(S)$ and $\{0, 1\}^S$ (functions $S \rightarrow \{0, 1\}$, of which there are $2^{|S|}$).

For infinite S the formula $|\mathcal{P}(S)| = 2^{|S|}$ still holds — and *Cantor’s theorem* (§8.3) asserts that $2^{|S|} > |S|$ even in the infinite case, which is the seed of the theory of higher cardinalities.